

Checkpoint 4: Residual Denoising Backbone

Objective

Replace the U-Net denoising backbone with a ResNet-style residual backbone under the same diffusion task.

Key Message

The proposed residual backbone improves structural and color-oriented metrics over the default baseline.

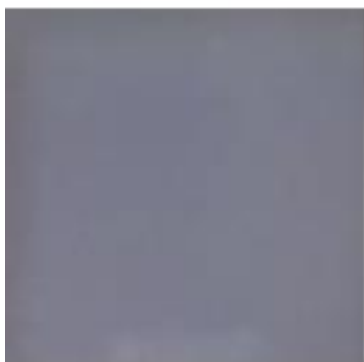
Completed Work

- Implemented residual blocks and residual denoising backbone.
- Trained the residual model using the same paired split and objective.
- Compared residual and baseline outputs on the same 368 test images.

Evidence

Metric	U-Net 128/50	Residual 128/50
MSE	0.039444	0.035727
PSNR	15.681501	16.304393
SSIM	0.610758	0.788963
Delta E	29.645203	26.840257

2.jpg turbid



2.jpg restored



2.jpg clear



8.jpg turbid



8.jpg restored



8.jpg clear



13.jpg turbid



13.jpg restored



13.jpg clear



26.jpg turbid



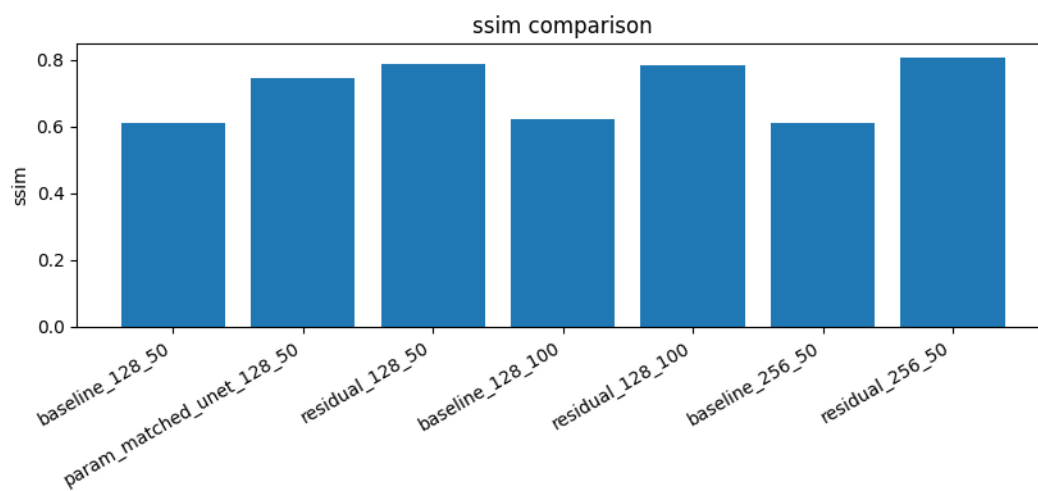
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Residual qualitative test examples.



SSIM comparison across experiments.

Notes

- Residual improves default-baseline SSIM by 29.18% and reduces Delta E by 9.46%.